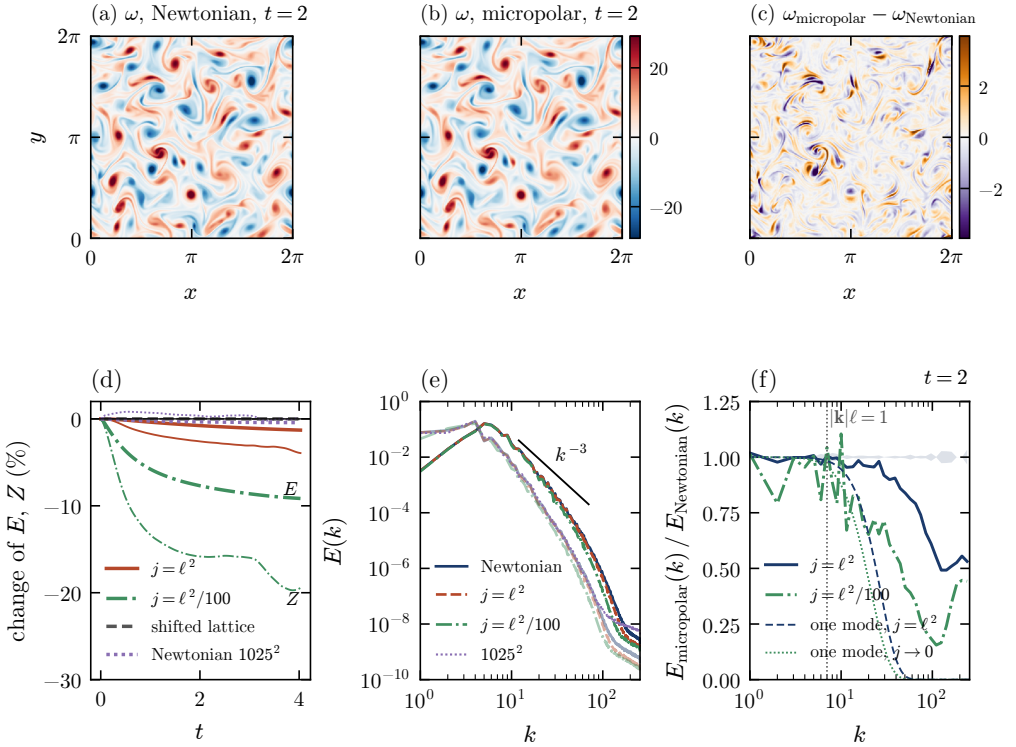

(a) ω, Newtonian, $t = 2$
(b) ω, micropolar, $t = 2$
(c) $\omega_{\text{micropolar}} - \omega_{\text{Newtonian}}$
2π
π
0
y
x
20
0
−20
2
0
−2
(d)
change of E, Z (%)
t
E
Z
$j = \ell^2$
$j = \ell^2/100$
shifted lattice
Newtonian 1025^2
(e)
$E(k)$
k
k^{-3}
Newtonian
$j = \ell^2$
$j = \ell^2/100$
1025^2
(f)
$t = 2$
$E_{\text{micropolar}}(k) \,/\, E_{\text{Newtonian}}(k)$
$k|\ell = 1$
$j = \ell^2$
$j = \ell^2/100$
one mode, $j = \ell^2$
one mode, $\gamma \to 0$